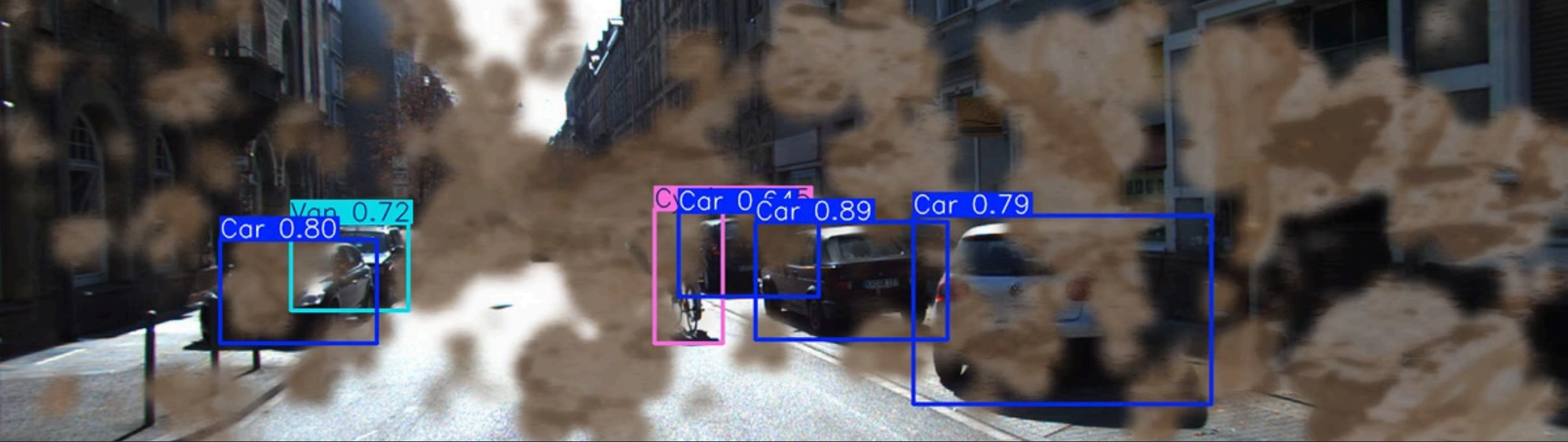




Robustifying ADAS Cameras Using Synthetic Augmentation

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Made with **GAMMA**

Review and Refine Project Definition

Motivation

ADAS fails with dirty lens. Standard datasets are "clean", leading to real-world performance gaps.

Project Definition

Develop a pipeline to augment datasets with realistic corruptions. Fine-tune object detection model for robustness.

Models

YOLOv8 (yolov8n.pt) for object detection.

Data

KITTI dataset transformed into "dirty" versions using synthetic overlays.

Metrics

Precision, Recall, F1 Score, Mean Confidence Score, mAP50.

Project Achievements and Novelty

Modular Synthetic Data Engine

Stable Diffusion generation of diverse "mud" and "water" overlays.



Automated Dirty Dataset

Applied varied overlays (intensity, position) to thousands of images.



Fine-Tuned YOLOv8

Outperformed standard pretrained models on corrupted footage.

“

A novel framework for **simulating complex dirty lens** artifacts via generative modeling, **eliminating the need for manual data acquisition** by synthesizing diverse edge-case datasets to optimize model reliability.

”

Review Methodology



Data Preparation

- 7,481 "dirty" images from KITTI dataset.
- Split: 70% Train, 20% Val, 10% Test.



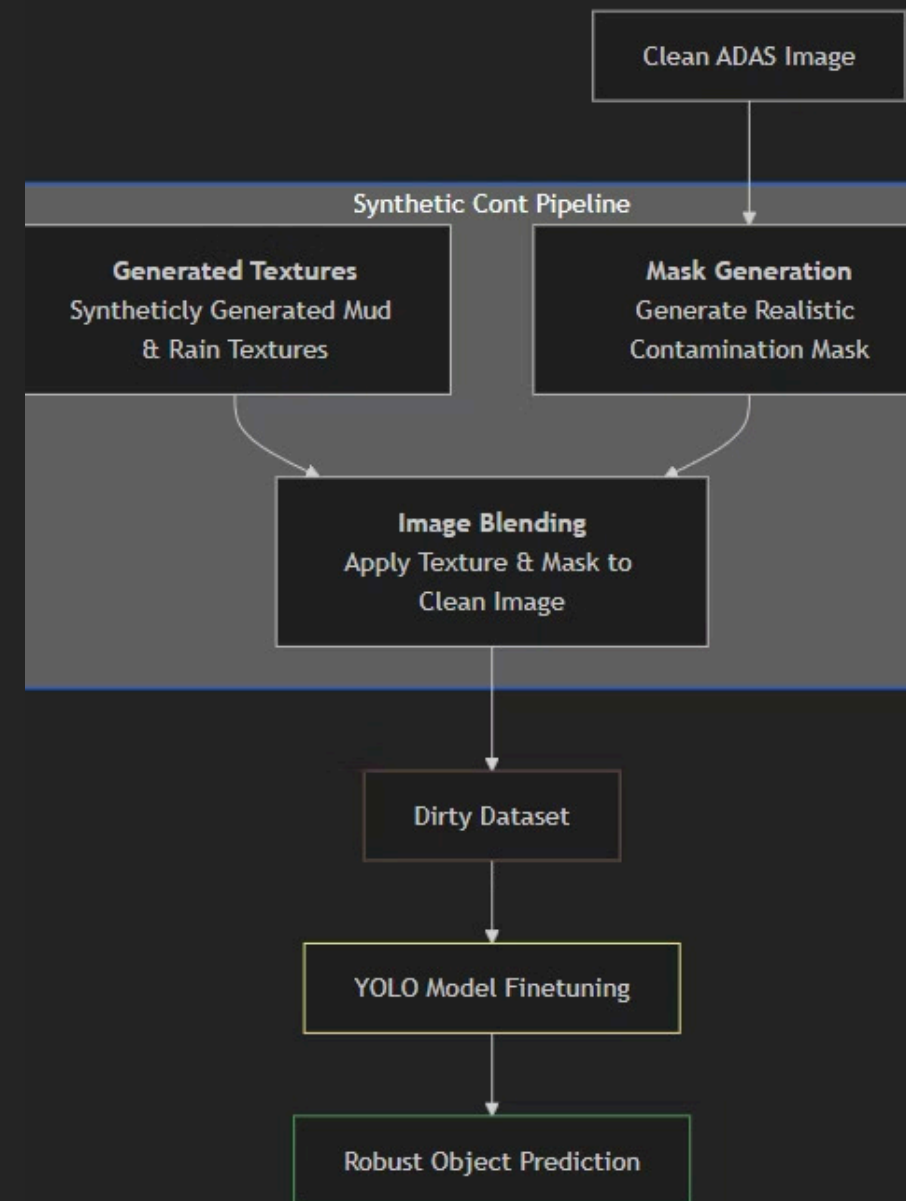
Model Training

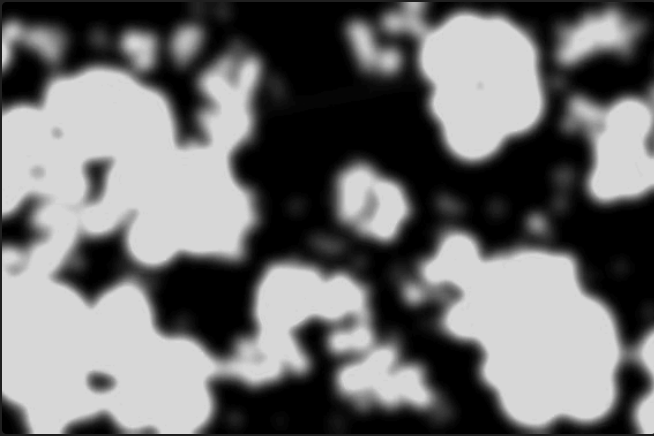
- Fine-tuned YOLOv8n on "dirty" set.
- Recognized objects like Cars, Trucks, Pedestrians and more.



Quality Effort

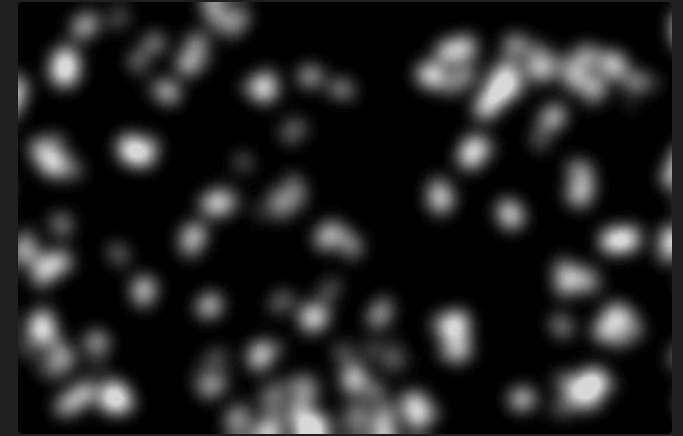
- **High-fidelity texture generation** - generating high quality, realistic contamination textures by carefully selecting the model and optimizing prompt design.
- **Realistic contamination mask creation** - designing and refining masks that accurately reflect real-world occlusion patterns, coverage, and spatial distribution
- **Dataset alignment with YOLO format** - resizing and fitting images and labels to match YOLO input resolution and annotation requirements while preserving spatial consistency.





1

A contamination mask is generated by randomly placing multi scale elliptical blobs followed by Gaussian smoothing to create continuous opacity transitions.



2

Realistic soiling textures are sampled from a pre-generated texture bank synthesized using Stable Diffusion, providing high-frequency, natural appearance details.

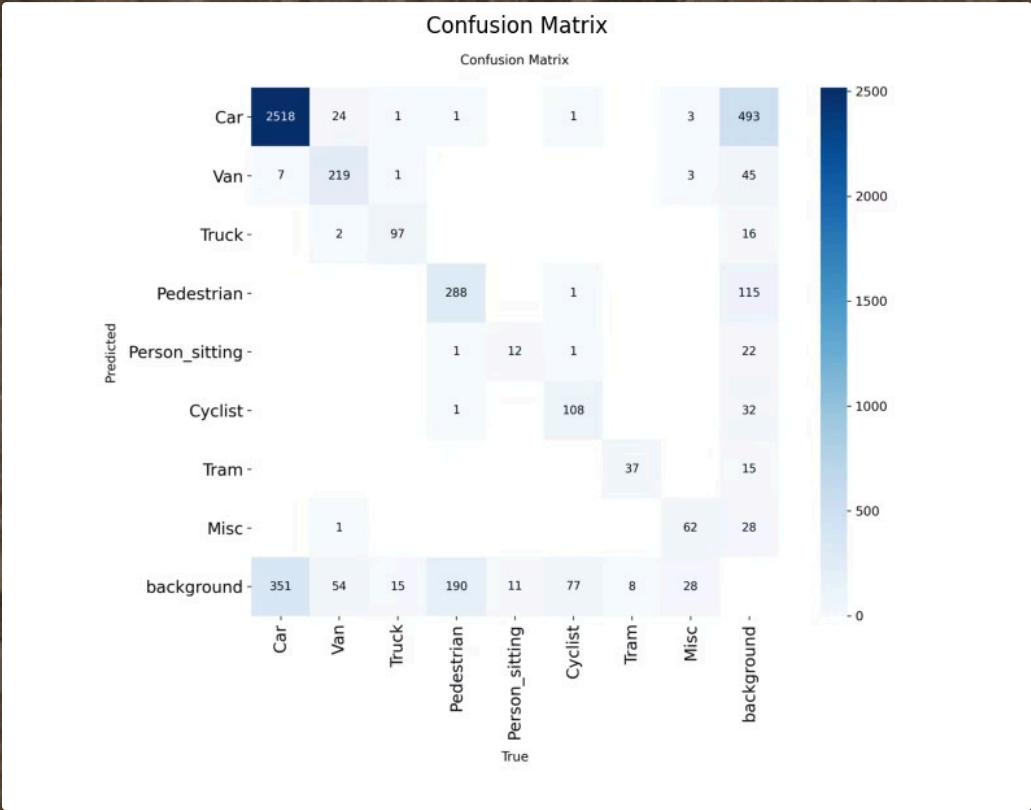
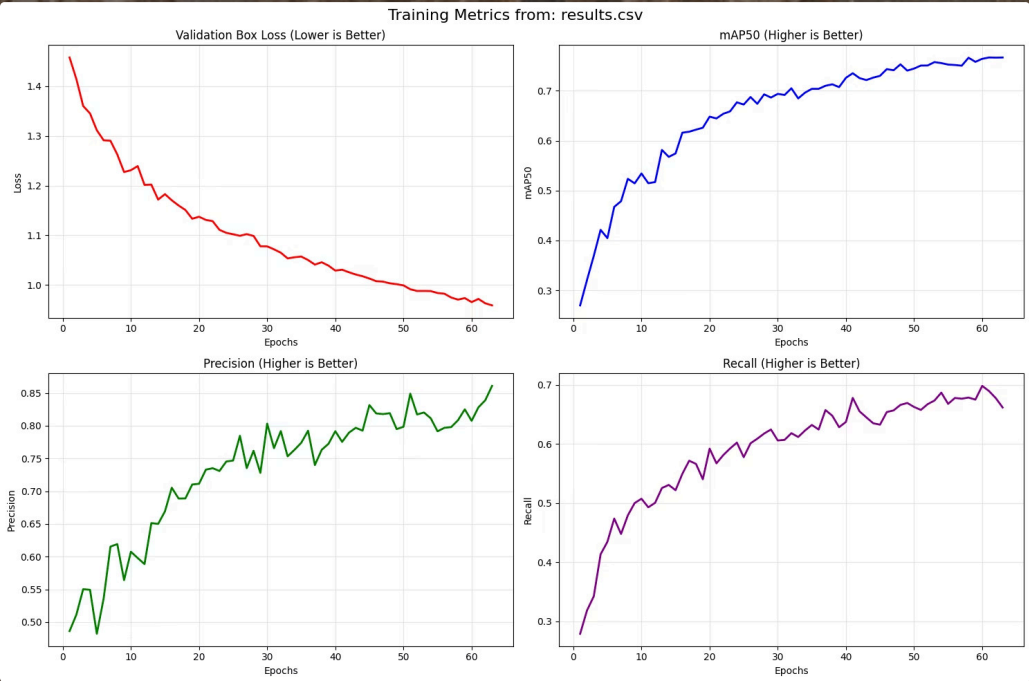


3

The final contaminated image is obtained by pixel-wise alpha blending and spatial filtering, where the mask modulates texture contribution and photometric degradation effects.

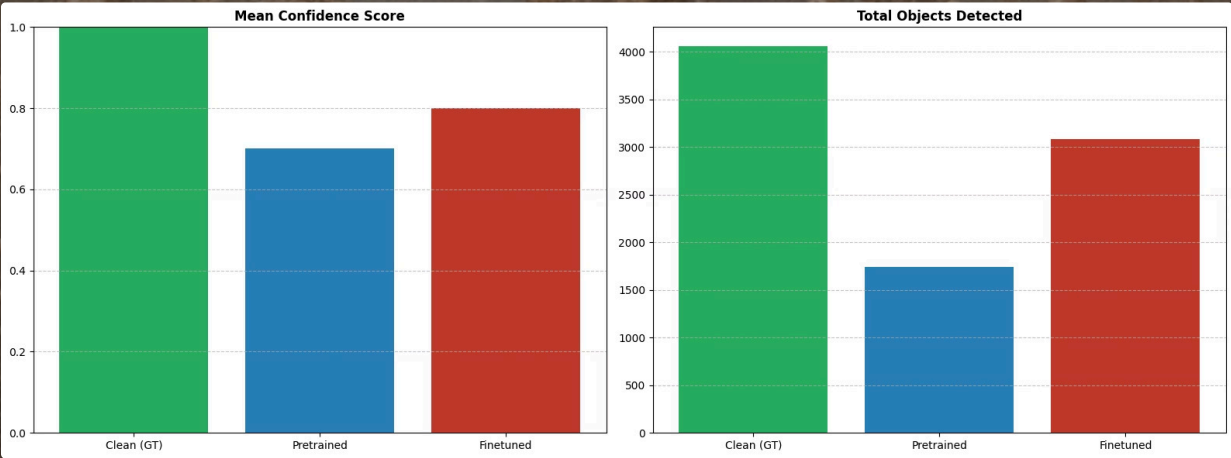


Results



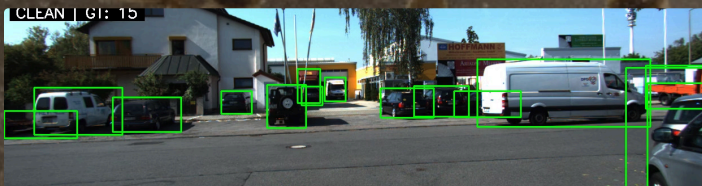
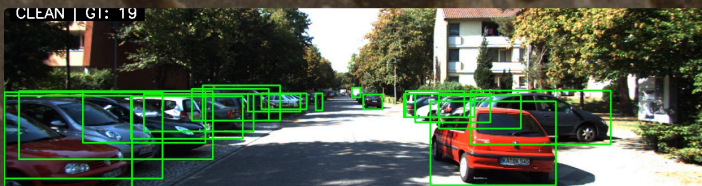
Pretrained Model

Precision: 0.5850
Recall: 0.5334
F1: 0.5294



Fine-tuned Model

Precision: 0.8413
Recall: 0.8403
F1: 0.8256



Conclusion



Desired Results Achieved

- **High-Fidelity Synthesis:** Developed an engine that generates realistic, varied dirt overlays.
- **Scalable Data:** Automated the creation of thousands of corrupted samples, **eliminating the manual effort** of physical data collection.
- **Proven Robustness:** Our fine-tuned **YOLOv8n** significantly outperformed standard models on corrupted ADAS footage.



Lessons Learned

- **Strategic Diversity:** Artifact variation (intensity/position) drives robustness more than raw volume.
- **Model & Prompt Selection:** Informed model selection and precise prompt engineering positively impacted the quality of the results.
- **Efficiency:** Synthetic generation eliminates the manual data acquisition bottleneck.

Future Ideas



Temporal Consistency

Extending the engine to video streams.



Real-World Validation

Test against actual dirty lens footage.



Broader Architectures

Incorporate more classes or Transformers.